LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Brief introduction to aircraft stability.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Brief introduction to aircraft stability. (Static and dynamic) explanation of longitudinal stability, lateral stability, directional stability
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams regarding stability axes; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of aircraft stability?
(b) List the three primary axes of aircraft motion.
(c) Define the term 'equilibrium' in the context of flight.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Static vs Dynamic Stability definitions • Longitudinal stability and CG location • Lateral stability and dihedral effect • Directional stability and vertical fin • Static margin and neutral point	L & Ds	PPT, IB	15 min
(b)	• Restoring moments in longitudinal axis • Rolling moments in lateral axis • Yawing moments in directional axis • Interrelationship of stability axes • Effect of speed on stability characteristics	L & Ds	PPT, PDF	20 min
(c)	• Role of AFCS in maintaining stability • Automatic stabilization mechanisms • Pilot inputs vs automatic corrections • Limitations of AFCS systems • Integration of stability with control	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the vertical fin contributes to directional stability.
(b) Describe the relationship between center of gravity and longitudinal stability.
(c) How does the AFCS assist the pilot in maintaining lateral stability?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of longitudinal, lateral, and directional stability discussed in this lesson will be expanded upon in LP 14 regarding divergence and Dutch roll.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to directional divergence, spiral divergence, Dutch roll and controllability.	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min